Lab 5: Static Code Analysis

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SECTION:H

BEFOR THE UPDATION:

1. Output of smart_home.py

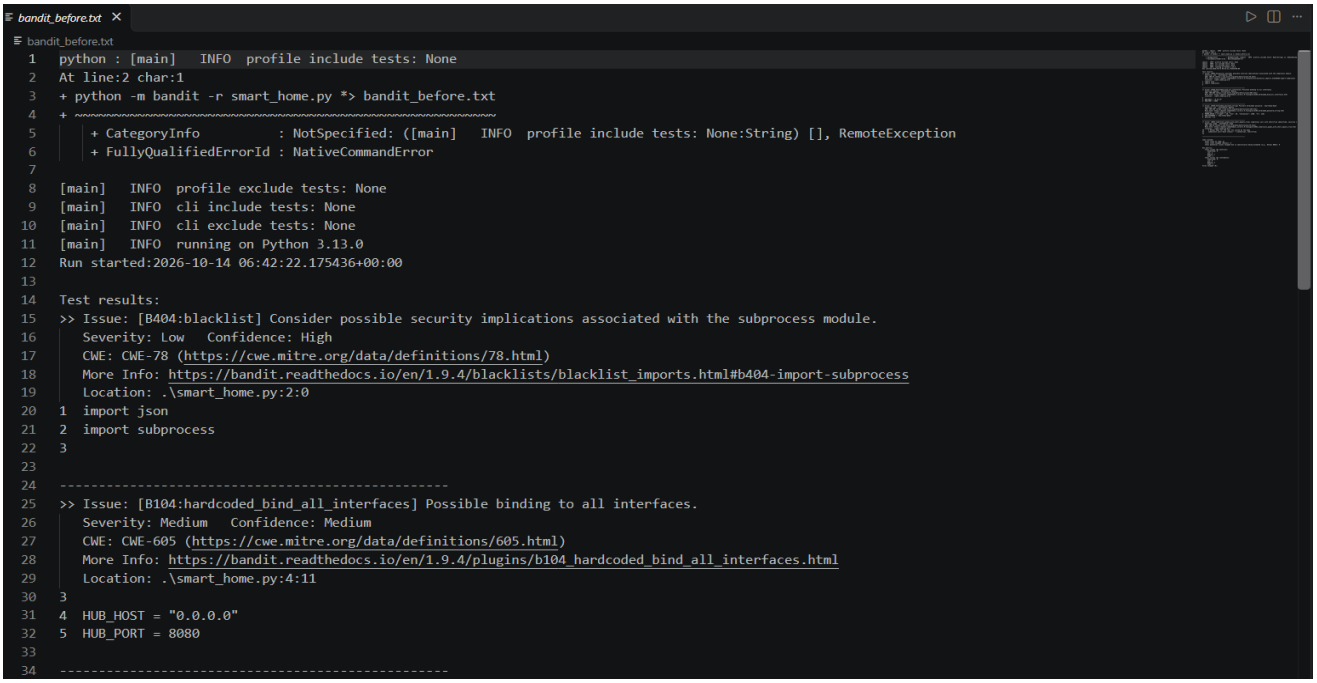
```
PS C:\Users\shiva\OneDrive\Desktop\SE\Lab05> python smart_home.py
Device not found: Garage Door
Ceiling Light is not a thermostat
Bedroom devices: ['Bedroom Fan', 'Thermostat']
Current power use (W): 1510
Hub address: 0.0.0.0:8080
Light settings: {'temperature': 30}
Devices on: ['Ceiling Light', 'Thermostat']
PS C:\Users\shiva\OneDrive\Desktop\SE\Lab05>
```

2. pylint_report.txt

pylint_before.txt

```
1  ************ Module smart_home
2  smart_home.py:1:0: C0114: Missing module docstring (missing-module-docstring)
3  smart_home.py:11:0: C0116: Missing function or method docstring (missing-function-docstring)
4  smart_home.py:11:0: C0103: Function name "addDevice" doesn't conform to snake_case naming style (invalid-name)
5  smart_home.py:11:14: C0103: Argument name "deviceName" doesn't conform to snake_case naming style (invalid-name)
6  smart_home.py:11:0: W0102: Dangerous default value {} as argument (dangerous-default-value)
7  smart_home.py:16:0: C0116: Missing function or method docstring (missing-function-docstring)
8  smart_home.py:16:0: C0103: Function name "toggleDevice" doesn't conform to snake_case naming style (invalid-name)
9  smart_home.py:16:17: C0103: Argument name "deviceName" doesn't conform to snake_case naming style (invalid-name)
10 smart_home.py:16:0: R1710: Either all return statements in a function should return an expression, or none of them should. (inconsistent-return-st
11 smart_home.py:23:0: C0116: Missing function or method docstring (missing-function-docstring)
12 smart_home.py:23:0: C0103: Function name "setTemperature" doesn't conform to snake_case naming style (invalid-name)
13 smart_home.py:23:19: C0103: Argument name "deviceName" doesn't conform to snake_case naming style (invalid-name)
14 smart_home.py:32:0: C0116: Missing function or method docstring (missing-function-docstring)
15 smart_home.py:32:0: C0103: Function name "devicesInRoom" doesn't conform to snake_case naming style (invalid-name)
16 smart_home.py:36:0: C0116: Missing function or method docstring (missing-function-docstring)
17 smart_home.py:36:0: C0103: Function name "powerUsage" doesn't conform to snake_case naming style (invalid-name)
18 smart_home.py:40:0: C0116: Missing function or method docstring (missing-function-docstring)
19 smart_home.py:40:0: C0103: Function name "saveState" doesn't conform to snake_case naming style (invalid-name)
20 smart_home.py:40:14: C0103: Argument name "fileName" doesn't conform to snake_case naming style (invalid-name)
21 smart_home.py:41:9: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
22 smart_home.py:45:0: C0116: Missing function or method docstring (missing-function-docstring)
23 smart_home.py:45:0: C0103: Function name "hubAddress" doesn't conform to snake_case naming style (invalid-name)
24 smart_home.py:49:0: C0116: Missing function or method docstring (missing-function-docstring)
25 smart_home.py:49:0: C0103: Function name "runScene" doesn't conform to snake_case naming style (invalid-name)
26 smart_home.py:49:13: C0103: Argument name "sceneScript" doesn't conform to snake_case naming style (invalid-name)
27 smart_home.py:54:0: C0116: Missing function or method docstring (missing-function-docstring)
28
29 ----------------------------------
30 Your code has been rated at 4.80/10
31
32
```

3. bandit_report.txt



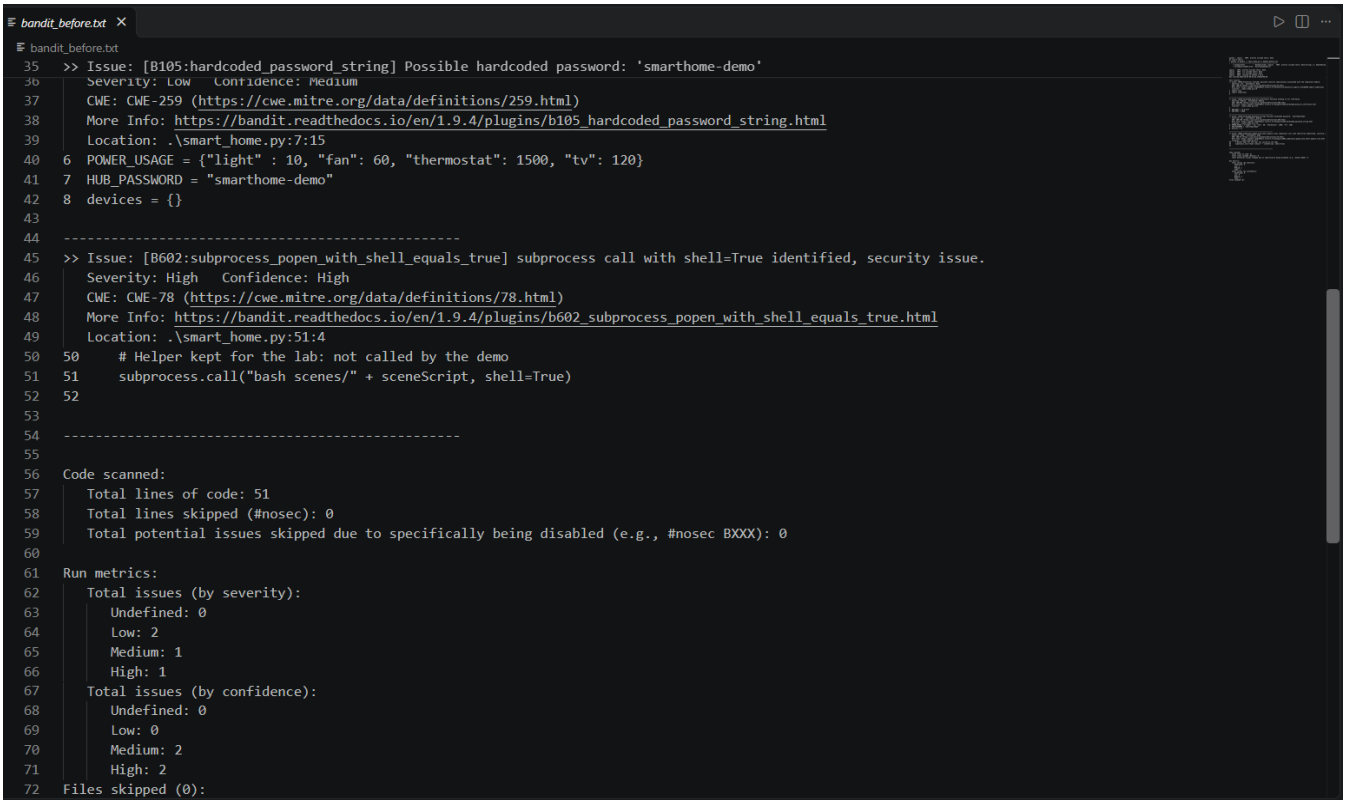
```
bandit_before.txt
python : [main]   INFO  profile include tests: None
At line:2 char:1
+ python -m bandit -r smart_home.py *> bandit_before.txt
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: ([main]   INFO  profile include tests: None:String) [], RemoteException
    + FullyQualifiedErrorId : NativeCommandError

[main]   INFO  profile exclude tests: None
[main]   INFO  cli include tests: None
[main]   INFO  cli exclude tests: None
[main]   INFO  running on Python 3.13.0
Run started:2026-10-14 06:42:22.175436+00:00

Test results:
>> Issue: [B404:blacklist] Consider possible security implications associated with the subprocess module.
   Severity: Low   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_imports.html#b404-import-subprocess
   Location: .\smart_home.py:2:0
1  import json
2  import subprocess
3

--------------------------------------------------
>> Issue: [B104:hardcoded_bind_all_interfaces] Possible binding to all interfaces.
   Severity: Medium   Confidence: Medium
   CWE: CWE-605 (https://cwe.mitre.org/data/definitions/605.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b104_hardcoded_bind_all_interfaces.html
   Location: .\smart_home.py:4:11
3
4  HUB_HOST = "0.0.0.0"
5  HUB_PORT = 8080

--------------------------------------------------
```



```
bandit_before.txt
>> Issue: [B105:hardcoded_password_string] Possible hardcoded password: 'smarthome-demo'
   Severity: Low   Confidence: Medium
   CWE: CWE-259 (https://cwe.mitre.org/data/definitions/259.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b105_hardcoded_password_string.html
   Location: .\smart_home.py:7:15
6  POWER_USAGE = {"light" : 10, "fan": 60, "thermostat": 1500, "tv": 120}
7  HUB_PASSWORD = "smarthome-demo"
8  devices = {}

--------------------------------------------------
>> Issue: [B602:subprocess_popen_with_shell_equals_true] subprocess call with shell=True identified, security issue.
   Severity: High   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b602_subprocess_popen_with_shell_equals_true.html
   Location: .\smart_home.py:51:4
50     # Helper kept for the lab: not called by the demo
51     subprocess.call("bash scenes/" + sceneScript, shell=True)
52

--------------------------------------------------

Code scanned:
   Total lines of code: 51
   Total lines skipped (#nosec): 0
   Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
   Total issues (by severity):
      Undefined: 0
      Low: 2
      Medium: 1
      High: 1
   Total issues (by confidence):
      Undefined: 0
      Low: 0
      Medium: 2
      High: 2
Files skipped (0):
```

4. flake8_report.txt

flake8_before.txt

```
smart_home.py:6:23: E203 whitespace before ':'
smart_home.py:40:1: E302 expected 2 blank lines, found 1
smart_home.py:66:40: E261 at least two spaces before inline comment
smart_home.py:68:80: E501 line too long (80 > 79 characters)
```

AFTER THE UPDATION :

1. Output of updated smart_home.py

```
PS C:\Users\shiva\OneDrive\Desktop\SE\Lab05> python smart_home_updated.py
Device not found: Garage Door
Ceiling Light is not a thermostat
Bedroom devices: ['Bedroom Fan', 'Thermostat']
Current power use (W): 1510
Hub address: 127.0.0.1:8080
Light settings: {}
Devices on: ['Ceiling Light', 'Thermostat']
PS C:\Users\shiva\OneDrive\Desktop\SE\Lab05>
```

2. pylint_report.txt

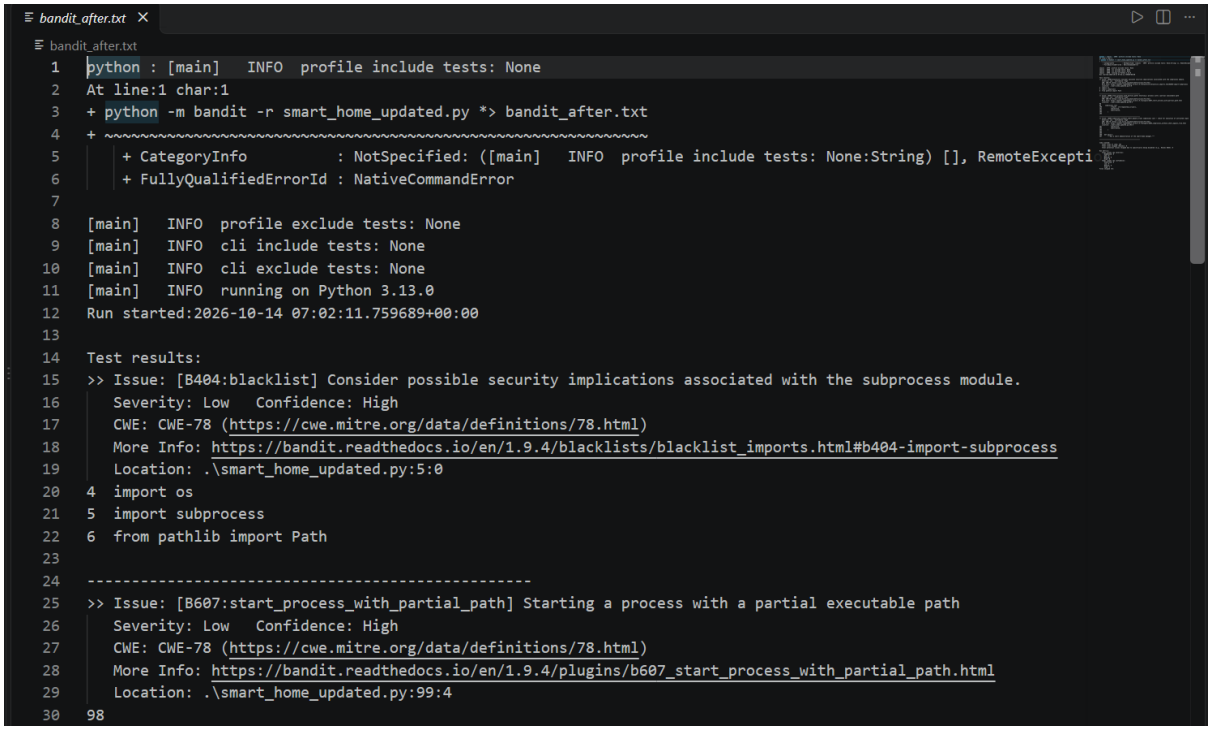
pylint_after.txt

```

---
Your code has been rated at 10.00/10


```

3. bandit_report.txt

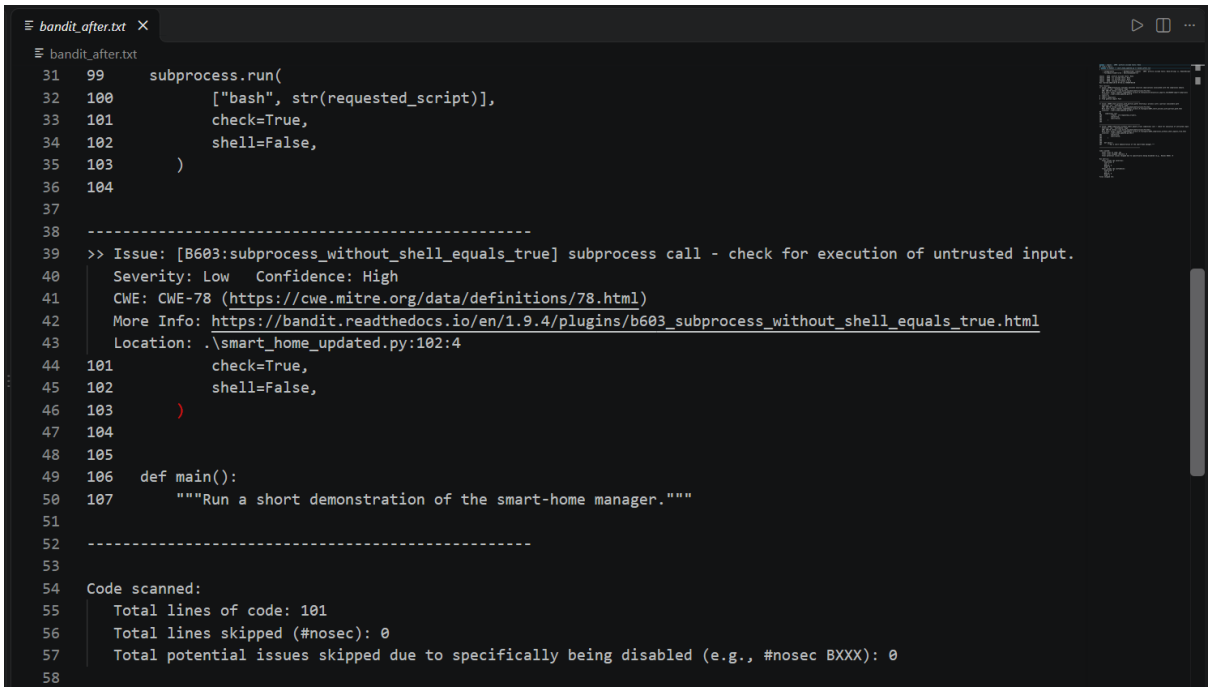


```
bandit_after.txt
python : [main]   INFO  profile include tests: None
At line:1 char:1
+ python -m bandit -r smart_home_updated.py *> bandit_after.txt
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: ([main]   INFO  profile include tests: None:String) [], RemoteExcepti
    + FullyQualifiedErrorId : NativeCommandError

[main]   INFO  profile exclude tests: None
[main]   INFO  cli include tests: None
[main]   INFO  cli exclude tests: None
[main]   INFO  running on Python 3.13.0
Run started:2026-10-14 07:02:11.759689+00:00

Test results:
>> Issue: [B404:blacklist] Consider possible security implications associated with the subprocess module.
   Severity: Low   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_imports.html#b404-import-subprocess
   Location: .\smart_home_updated.py:5:0
4  import os
5  import subprocess
6  from pathlib import Path

--------------------------------------------------
>> Issue: [B607:start_process_with_partial_path] Starting a process with a partial executable path
   Severity: Low   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b607_start_process_with_partial_path.html
   Location: .\smart_home_updated.py:99:4
98
```



```
bandit_after.txt
99      subprocess.run(
100             ["bash", str(requested_script)],
101             check=True,
102             shell=False,
103         )
104

--------------------------------------------------
>> Issue: [B603:subprocess_without_shell_equals_true] subprocess call - check for execution of untrusted input.
   Severity: Low   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b603_subprocess_without_shell_equals_true.html
   Location: .\smart_home_updated.py:102:4
101             check=True,
102             shell=False,
103         )
104

105
106     def main():
107         """Run a short demonstration of the smart-home manager."""

--------------------------------------------------

Code scanned:
   Total lines of code: 101
   Total lines skipped (#nosec): 0
   Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
   Total issues (by severity):
      Undefined: 0
      Low: 3
      Medium: 0
      High: 0
   Total issues (by confidence):
      Undefined: 0
      Low: 0
      Medium: 0
      High: 3
Files skipped (0):
```

4. flake8_report.txt

